OF-3 Report :
Simulating Supersonic Flow over a step

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2.1 Two-dimensional Euler equations

We consider the Euler equations in $\mathbb{R}^2$ complemented by the equation of state of an ideal gas (Eq. 6) with the specific heat ratio $\gamma = 1.4$. The two-dimensional Euler equations read:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x} + \frac{\partial \mathbf{G}}{\partial y} = 0 \tag{1}$$

Where $\mathbf{U}$ is the state vector, $\mathbf{F}$ is the x-flux vector, and $\mathbf{G}$ is the y-flux vector.

From the assumptions of Euler equations, e.g., no body forces and inviscid and isothermal flow, the simplified form *Navier-Stokes Equations* read:

- Continuity :

$$\frac{\partial \rho}{\partial t} + \rho \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) = 0 \tag{2}$$

- X-Momentum :

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \rho \left(\frac{\partial u^2}{\partial x} + \frac{\partial uv}{\partial y} \right) = -\frac{\partial p}{\partial x} \tag{3}$$

- Y-Momentum :

$$\frac{\partial \rho \mathbf{v}}{\partial t} + \rho \left(\frac{\partial uv}{\partial x} + \frac{\partial v^2}{\partial y} \right) = -\frac{\partial p}{\partial y} \tag{4}$$

- Energy (inviscid simplification):

$$\frac{\partial E}{\partial t} + E \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) = -p \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) \tag{5}$$

  Where $\underline{E = \rho(e + ||\boldsymbol{u}||^2/2)}$. e is the internal energy of the fluid per unit mass (Joule per kilogram in SI).

Thus the state vector $\mathbf{U} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho(e + \frac{1}{2}(u^2 + v^2)) \end{bmatrix} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ E \end{bmatrix}$.

The x-flux vector $\mathbf{F} = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ (\rho(e + \frac{1}{2}(u^2 + v^2)) + p) * u \end{bmatrix} = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ (E + p)(u) \end{bmatrix}$

The y-flux vector $\mathbf{G} = \begin{bmatrix} \rho v \\ \rho vu \\ \rho v^2 + p \\ (\rho(e + \frac{1}{2}(u^2 + v^2)) + p)(v) \end{bmatrix} = \begin{bmatrix} \rho v \\ \rho vu \\ \rho v^2 + p \\ (E + p)(v) \end{bmatrix}$

The *Ideal Gas Law* is our equation of state that relates the variables, ρ which is our fluids density, p the local pressure, R the molecular gas constant, and T the local temperature.

$$p = \rho RT \tag{6}$$

Using the first law of Thermodynamics, p can be related to the specific internal energy e and the specific heat ratio γ by:

$$e = \frac{p}{\rho(\gamma - 1)} \tag{7}$$

These can be combined for a new expression that expresses p entirely in terms of the state vector $\mathbf{U}$, allowing the flux vectors $\mathbf{F}(\mathbf{U})$ and $\mathbf{G}(\mathbf{U})$ to be evaluated directly from conserved quantities.

$$p = (\gamma - 1)\left(E - \frac{1}{2}\rho(u^2 + v^2)\right) \tag{8}$$

2.2 Local Sound Speed and Mach Number

To analyze compressible flow behavior, we often need the local sound speed and Mach number.

Local Sound Speed:
The local speed of sound a in a compressible fluid is given by the equation:

$$a = \sqrt{\gamma RT} = \sqrt{\gamma \frac{p}{\rho}} \tag{9}$$

Local Mach Number:
The local Mach number M is defined as the ratio of the flow speed to the local speed of sound:

$$M = \frac{\sqrt{u^2 + v^2}}{a} = \frac{|\mathbf{U}|}{a} \tag{10}$$

where u and v are the flow velocity components in the x and y directions, respectively.

2.3 Normal Shock and Isentropic Flow Relations

Here we look at the equations for normal shocks and isentropic flows,i.e., adiabatic and reversible.

For a normal shock with upstream (region 1) Mach number M_1 and downstream (region 2) Mach number M_2, the following standard jump conditions apply (for a perfect gas with ratio of specific heats γ):

- **Static Pressure Ratio:**

  $$\frac{p_2}{p_1} = 1 + \frac{2\,\gamma}{\gamma + 1}\big(M_1^2 - 1\big). \tag{11}$$

  where:

  - p_1: upstream (pre-shock) static pressure,

– p_2: downstream (post-shock) static pressure.

- **Downstream Mach Number:**

$$M_2^2 = \frac{1 + \frac{\gamma-1}{2} M_1^2}{\gamma M_1^2 - \frac{\gamma-1}{2}}. \tag{12}$$

Once the flow is on either side of a shock (and assuming no further shocks/viscous effects in that region), it can be treated as isentropic. In an isentropic flow of a perfect gas:

- **Isentorpic Pressure Relation:**

$$p_0 = p \cdot (1 + \frac{(\gamma - 1)}{2} M_1^2)^{\frac{\gamma}{\gamma-1}} \tag{13}$$

where:

– p_0: total (stagnation) pressure,

– p: static pressure at a given point in the flow,

– M: local Mach number at that point.

2.4 Step Example

Considering a flow over our step at $M = 3$ and $p = 1$ Pa, what is the pressure at the stagnation point (x=0.6m, y=0m in Figure 1) on the surface of the step? We will employ the equations laid out above.

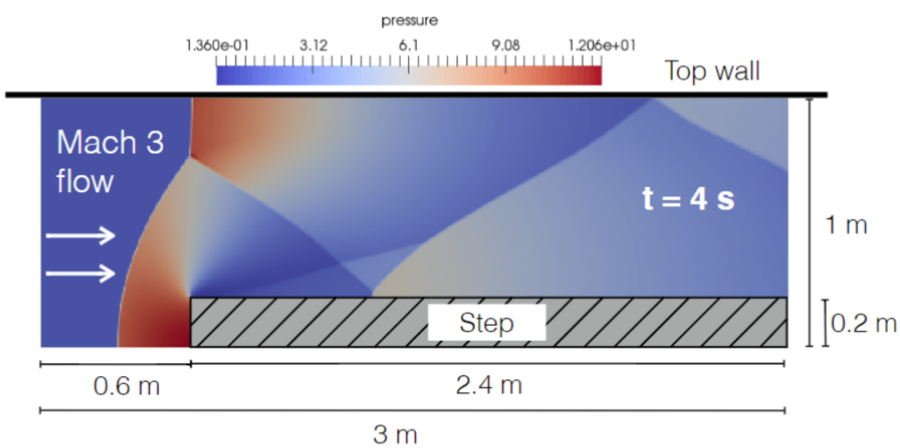
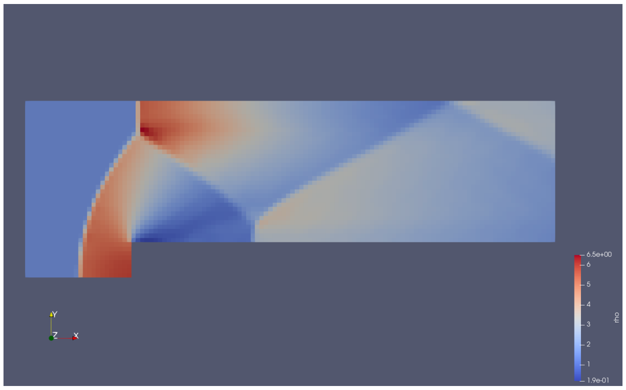


Figure 1: Overview of the Flow

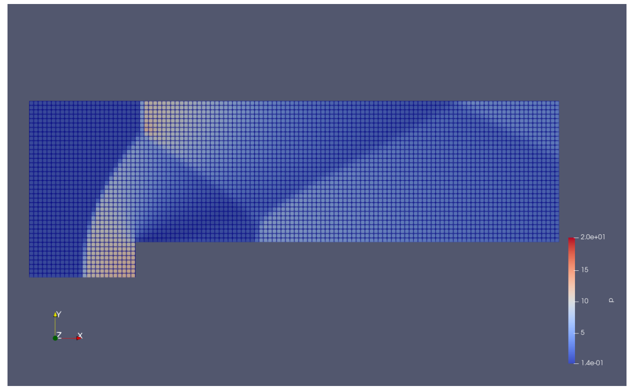
Using Equation 13 we can obtain the total pressure in the first region $p_{0_1} = 36.73$ Pa. We can also obtain the local Mach past the shock using Equation 12, $M_2 = 0.4752$. We can obtain our static pressure past the shock as $\boxed{p_2 = p_1 * \text{Eq. } 11 = 10.33 \text{ Pa}}$.

3 First Solution

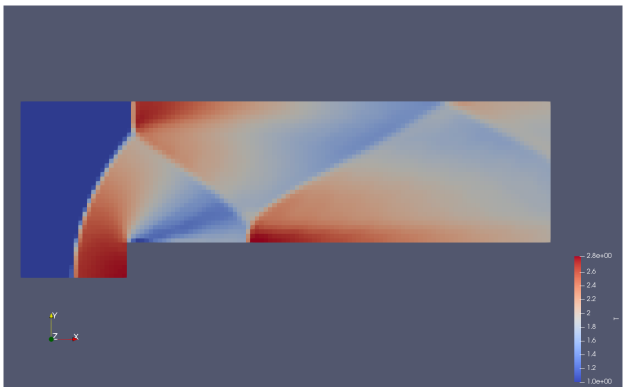
In this section we visualize the properties pressure p, density ρ, velocity $\mathbf{U}$, and temperature T after simulating Mach 3 flow over the step for $T = 4$ s.



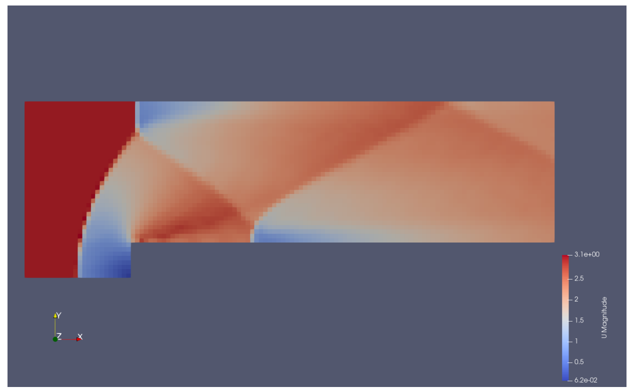
(a) Density ρ



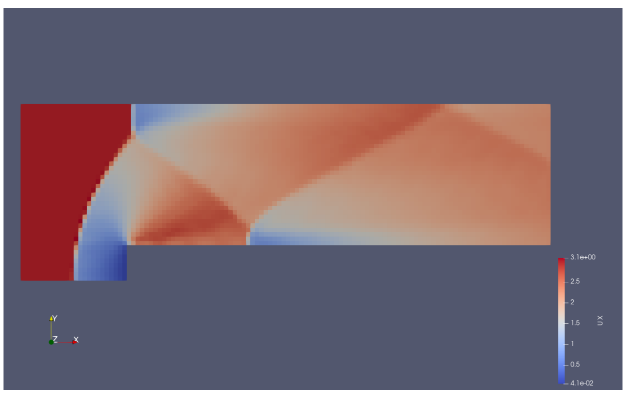
(b) Pressure p



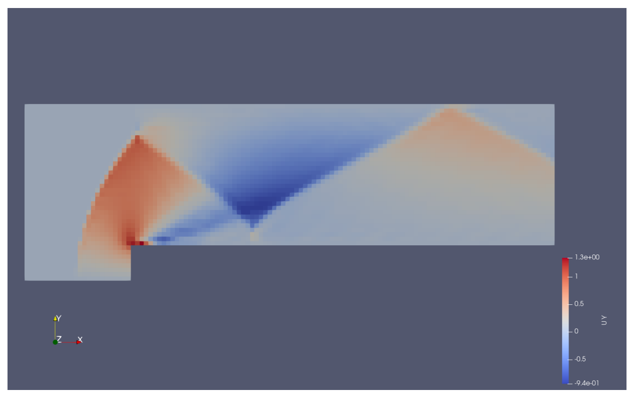
(c) Temperature T



(d) Magnitude of Velocity $|\mathbf{U}|$



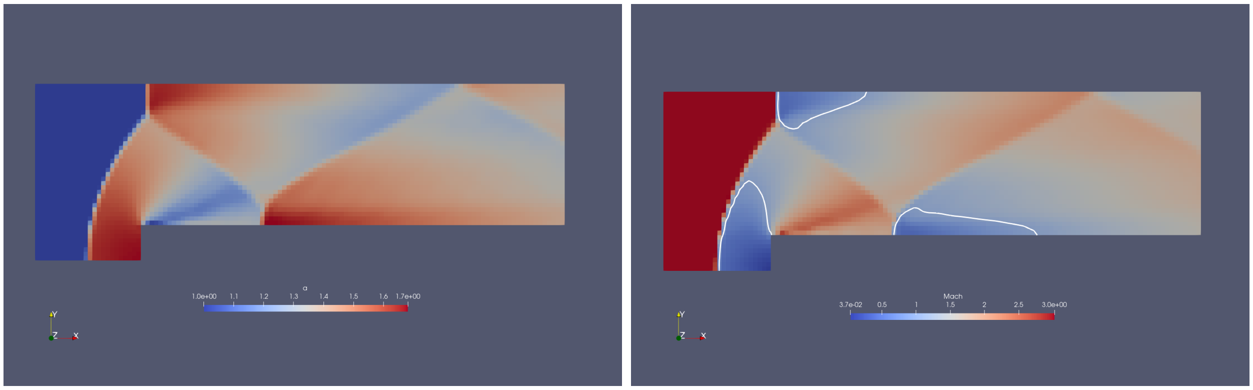
(e) x-velocity u



(f) y-velocity v

Figure 2: Coarse mesh solution, gridpoints can be seen in the pressure plot

Using the built-in calculator in *Paraview*, we are able to employ Equations 9 and 10 to visulaize the Mach and Speed of Sound in our supersonic simulation.



(a) Speed of Sound a

(b) Mach No. M

Figure 3: White countour lines representing sonic flow, i.e., where $M = 1$

The pressure at the stagnation grid point in this coarse mesh is $p = 11.9647$ Pa. This is close but greater than what has been computed using our normal shock theory.

4 Converging the Solution

5 Parallel Solver Performance

Appendix

A Code

PDF of code starts on next page.

Additionally, team repository: `https://github.com/andressuniaga/CFD-SP25-11`